

Day 2 – 3,4



Multimodal models

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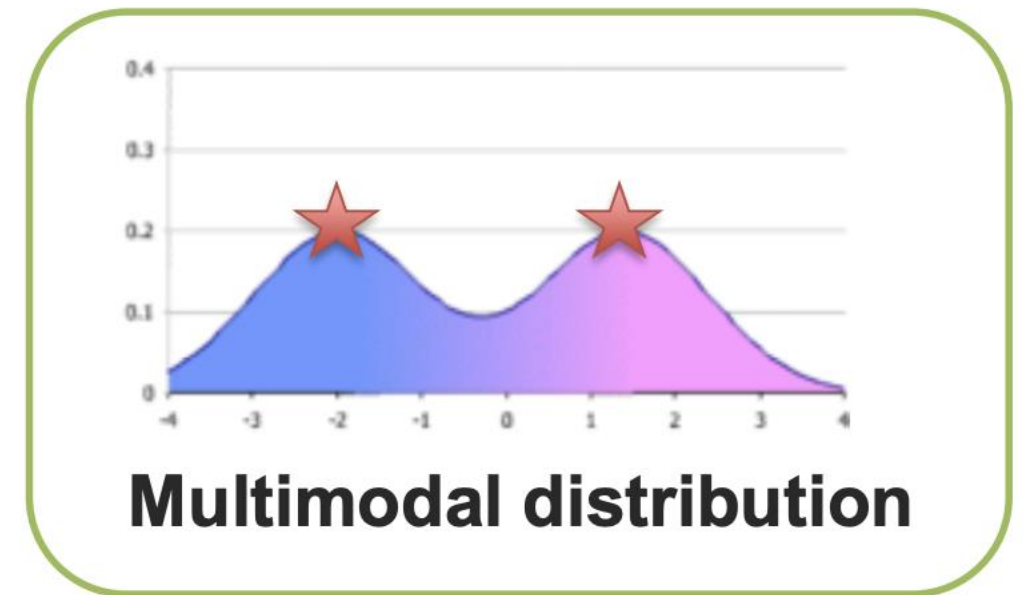
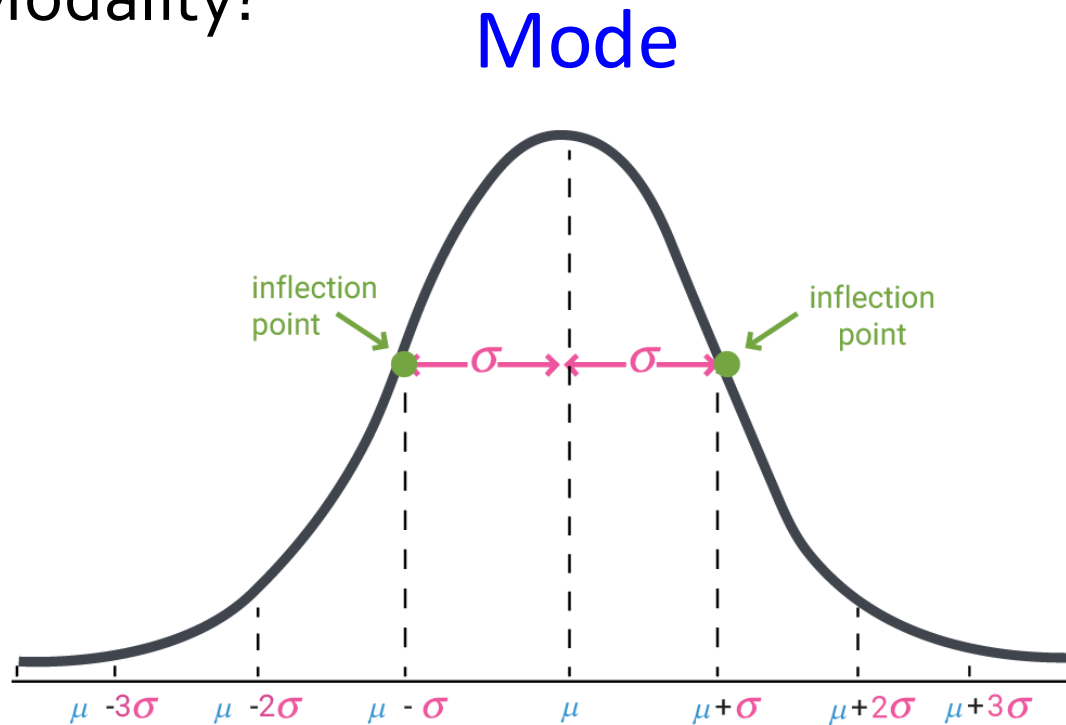
Schedule

Each hour is scheduled as **50 min lecture + 10 min break**

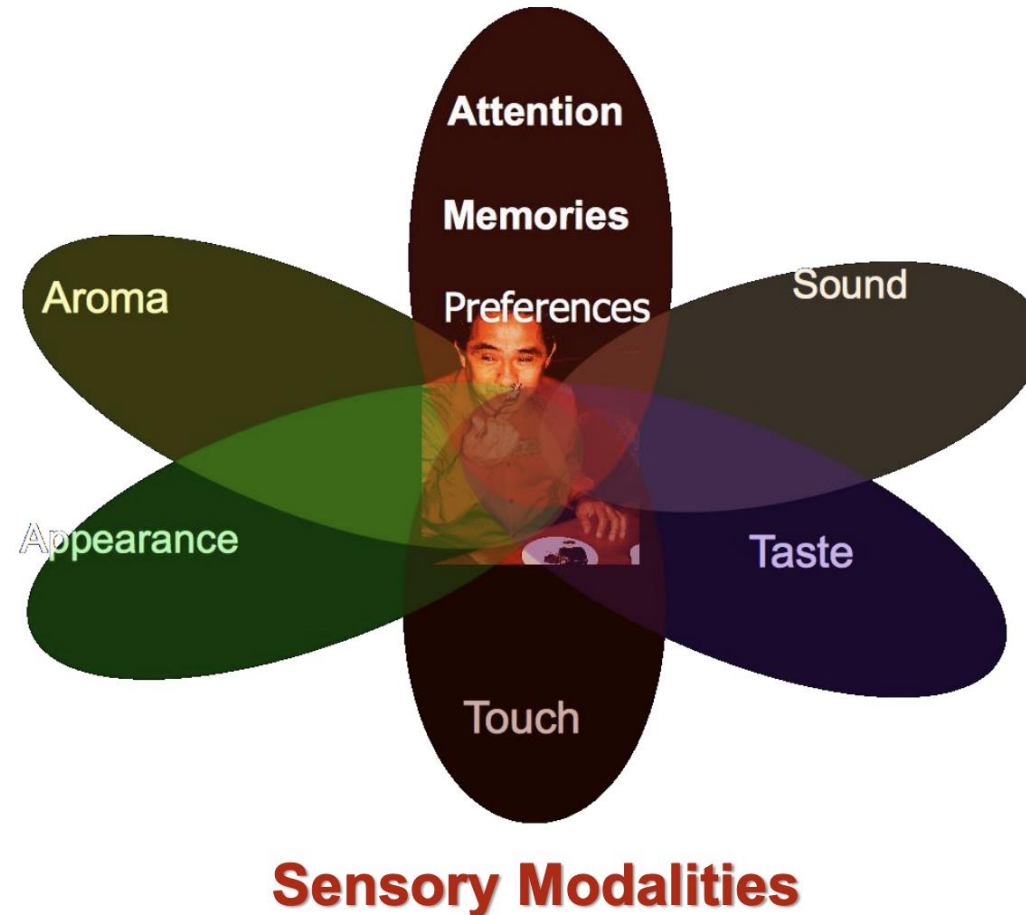
- **Day 1:** 4 hr lecture + 4 hr practice
 - **AM:** CNN, Transformers, Object detection
 - **PM:** DETR, MI-DETR
- **Day 2:** 4 hr lecture + 4 hr practice
 - **AM:** Self-supervised learning, Weakly supervised learning, Multimodal models, Semantic segmentation
 - **PM:** DINO, SAM variants

What is multi-modal recognition?

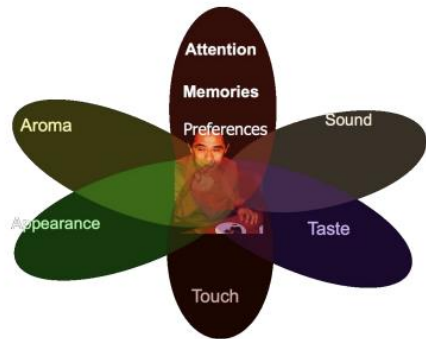
- Recognition under multiple modalities
- Modality?



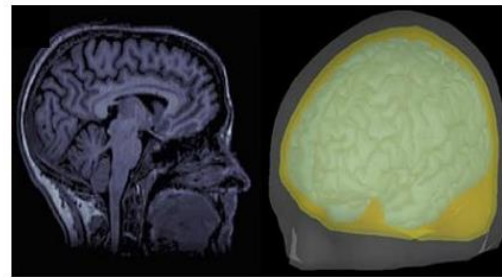
What is multi-modal recognition? (cont'd)



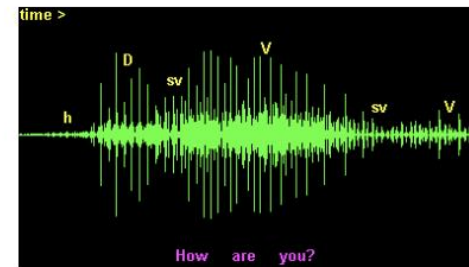
Applications of multi-modal recognition



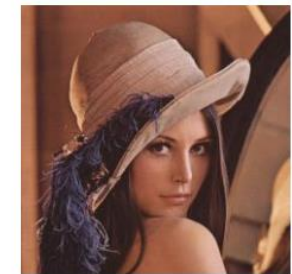
Psychology



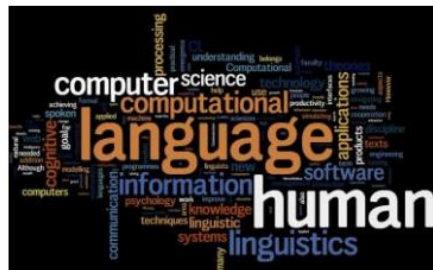
Medical



Speech



Vision



Language



Multimedia



Robotics

A photograph of a whiteboard filled with mathematical equations, representing learning.

Learning

Multimodal GenAI upto 2023

DALL-E3 (OpenAI) [1]

A swirling, multicolored portal emerges from the depths of an ocean of coffee, with waves of the rich liquid gently rippling outward. The portal engulfs a coffee cup, which serves as a gateway to a fantastical dimension. The surrounding digital art landscape reflects the colors of the portal, creating an alluring scene of endless possibilities.



Make-A-Video (Meta) [2]



Phenaki (Google) [3]



[1] Betker *et al.*, **Improving Image Generation with Better Captions**, 2023 (a.k.a. [DALL-E3](#) from OpenAI)

[2] Singer *et al.*, **Make-A-Video: Text-to-Video Generation without Text-Video Data**, *ICLR 2023* (Meta AI)

[3] Villegas *et al.*, **Phenaki: Variable Length Video Generation from Open Domain Textual Descriptions**, *ICLR 2023* (Google Brain)

Sora (Feb. 2024)

Input prompt: The camera directly faces colorful buildings in Burano Italy. An adorable dalmation looks through a window on a building on the ground floor. Many people are walking and cycling along the canal streets in front of the buildings.

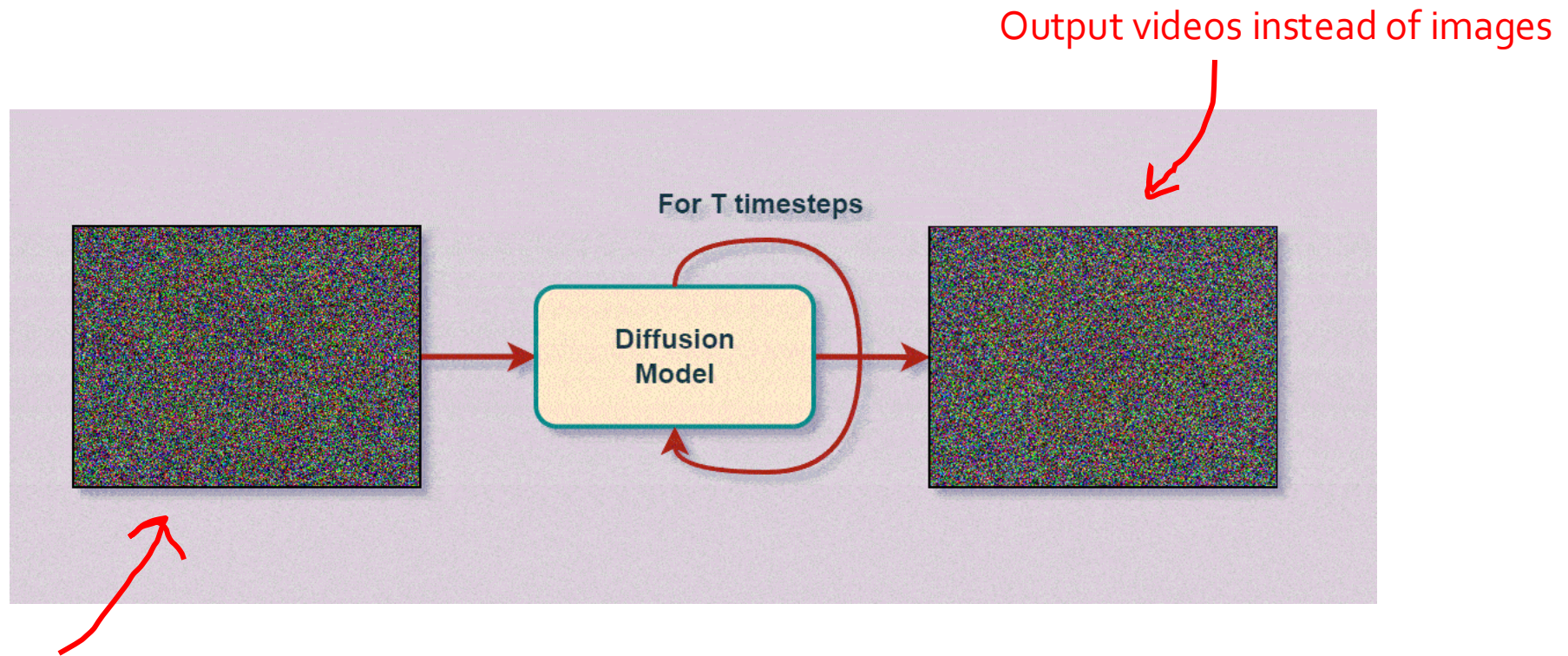


Gen-4 (March 2025)



By Runway

CLIP + Diffusion

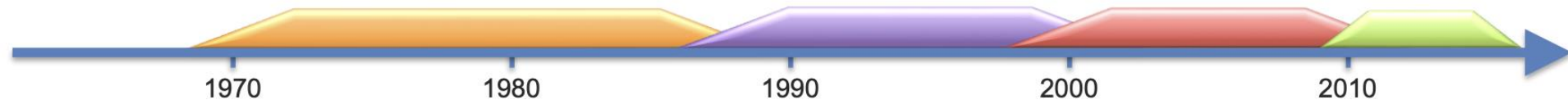


This random vector is mapped with CLIP embedding

Prior research on “Multimodal”

Four eras of multimodal research

- The “**behavioral**” era (1970s until late 1980s)
- The “**computational**” era (late 1980s until 2000)
- The “**interaction**” era (2000 - 2010)
- The “**deep learning**” era (2010s until ...)



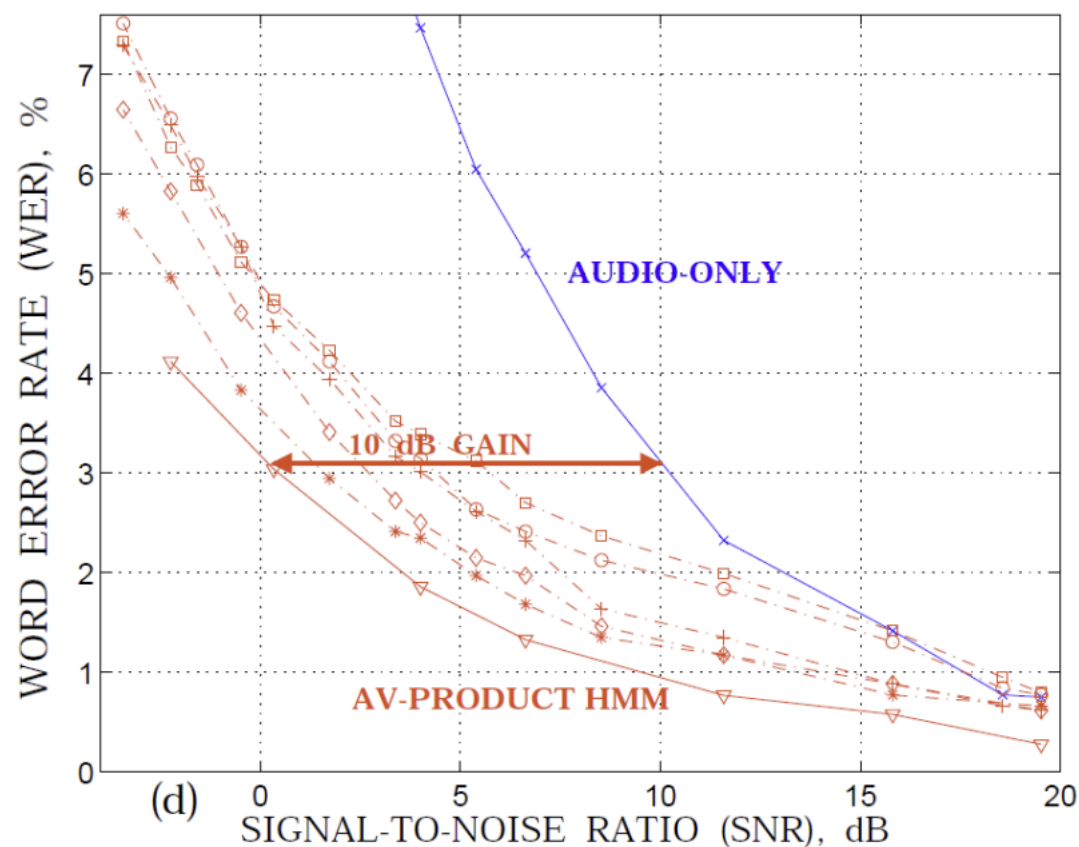
Multimodal interference: The McGurk Effect (1976)



McGurk and MacDonald "Hearing lips and seeing voices" *Nature* 1976

The “Computational” era (Late 1980s – 2000)

- Audio-Visual Speech Recognition (AVSR)



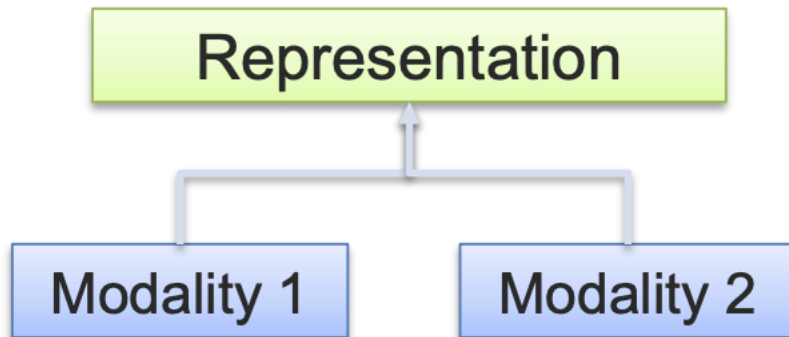
The “Deep Learning” era (2010-Now)

- Core Challenges in “Deep” Multimodal ML
 - Representation
 - Alignment
 - Fusion
 - Translation
 - Co-Learning

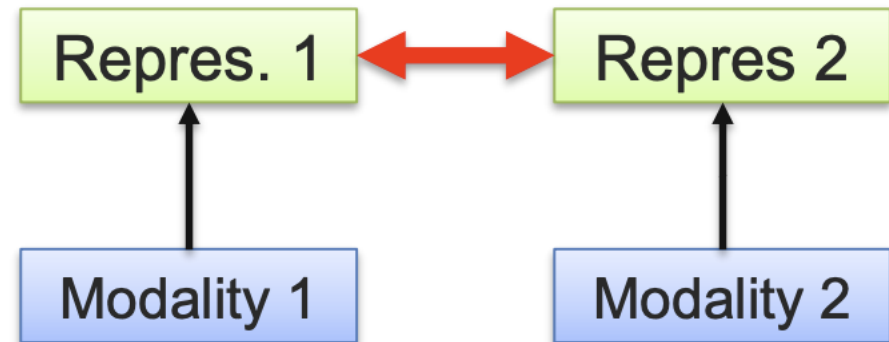
Core Challenge 1: Representation

- **Definition:** Learning how to represent and summarize multimodal data in a way that exploits the complementarity and redundancy.

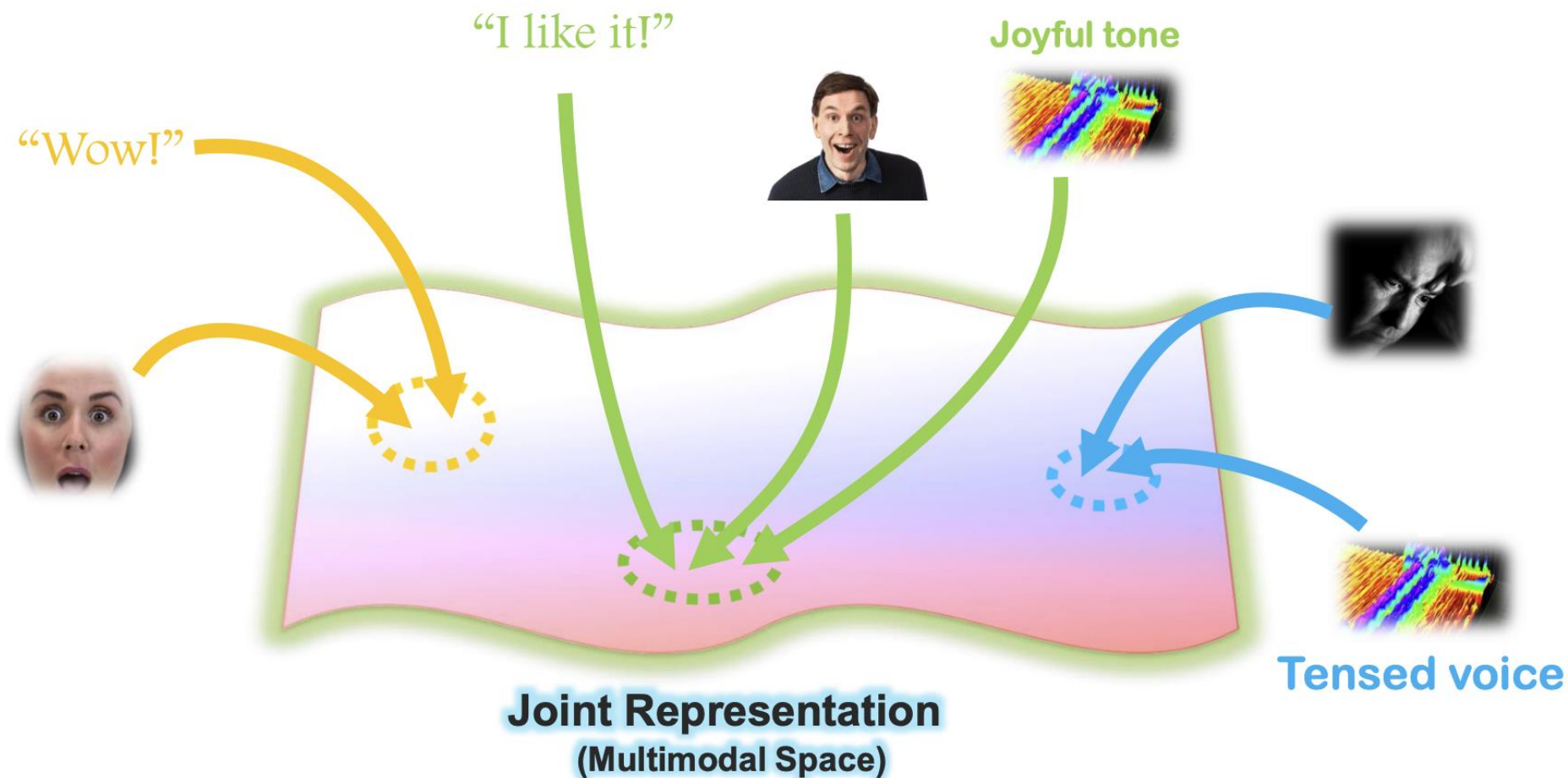
Ⓐ Joint representations:



Ⓑ Coordinated representations:



Joint multimodal representation



Other examples of joint multimodal representation

Audio-visual speech recognition

[Ngiam et al., ICML 2011]

- Bimodal Deep Belief Network

Image captioning

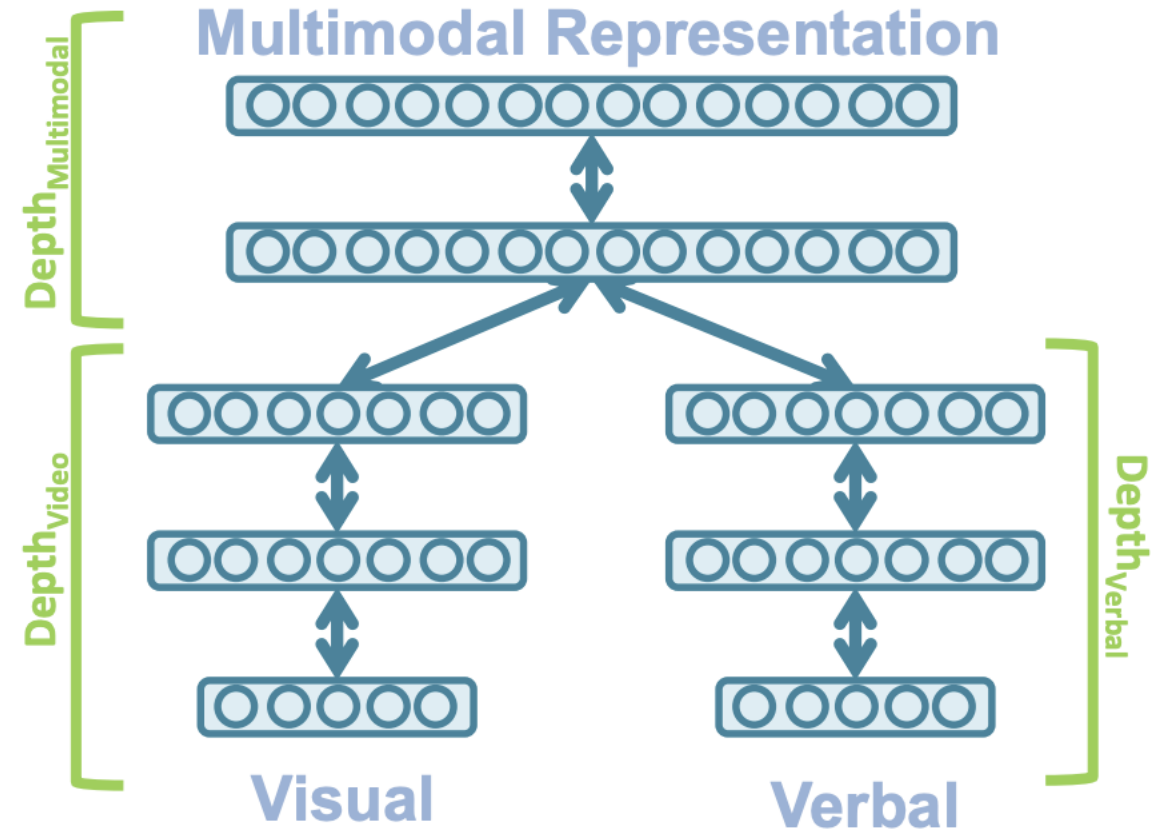
[Srivastava and Salahutdinov, NIPS 2012]

- Multimodal Deep Boltzmann Machine

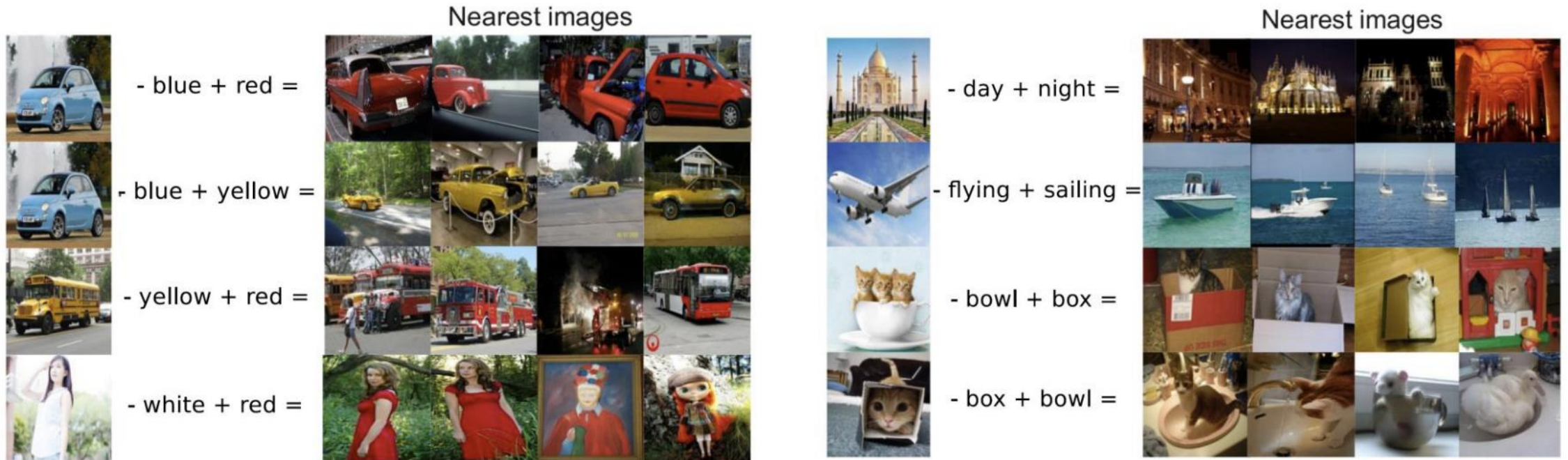
Audio-visual emotion recognition

[Kim et al., ICASSP 2013]

- Deep Boltzmann Machine



Multimodal vector space arithmetic

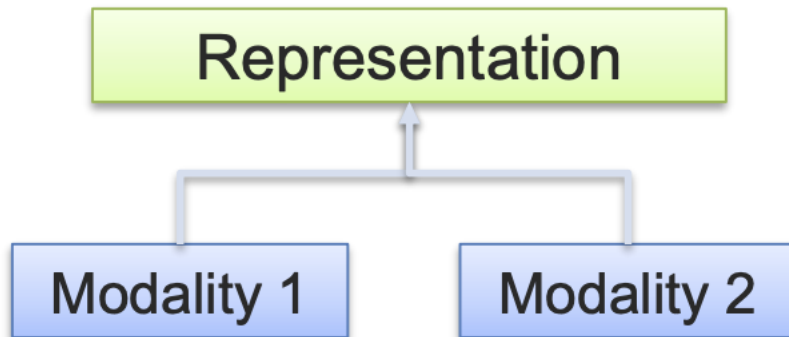


[Kiros et al., Unifying Visual-Semantic Embeddings with Multimodal Neural Language Models, 2014]

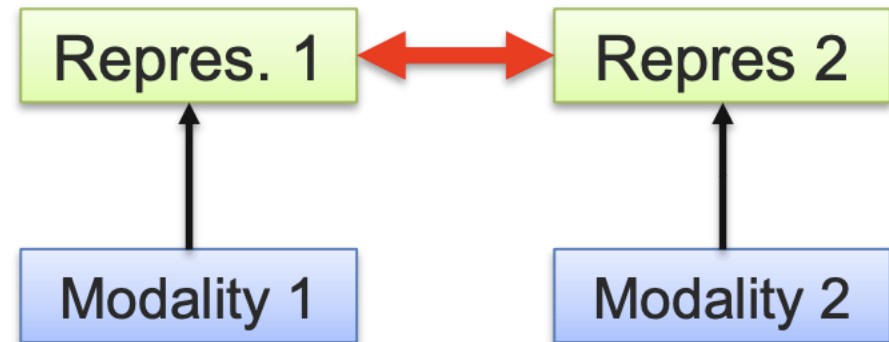
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Ⓐ Joint representations:

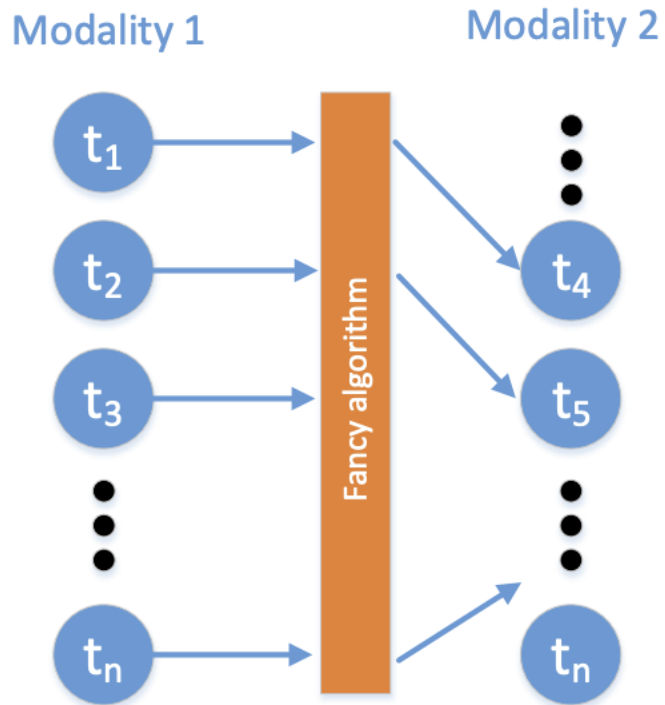


Ⓑ Coordinated representations:



Core challenge 2: Alignment

- **Definition:** Identify the direct relations between (sub)elements from two or more different modalities



A Explicit Alignment

The goal is to directly find correspondences between elements of different modalities

B Implicit Alignment

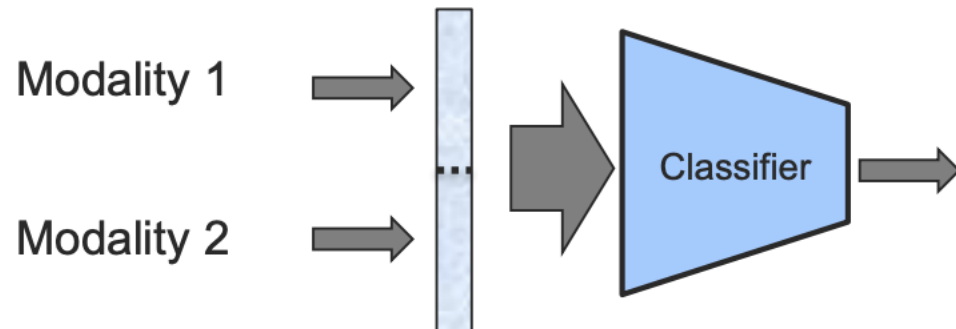
Uses internally latent alignment of modalities in order to better solve a different problem

Core challenge 3: Fusion

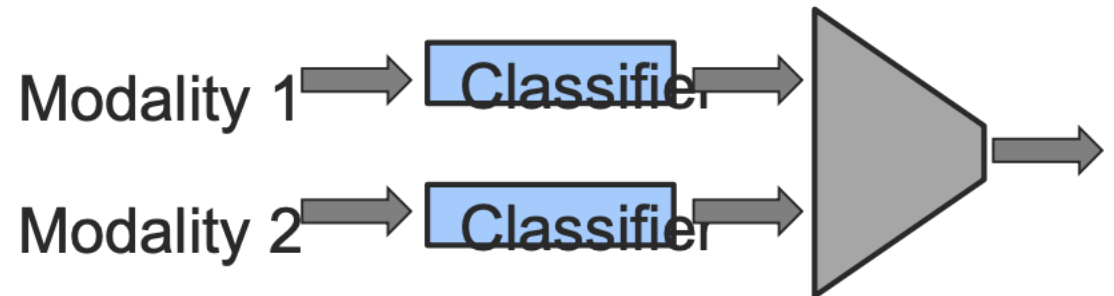
- **Definition:** Join information from two or more modalities to perform a prediction task.

A Model-Agnostic Approaches

1) Early Fusion



2) Late Fusion

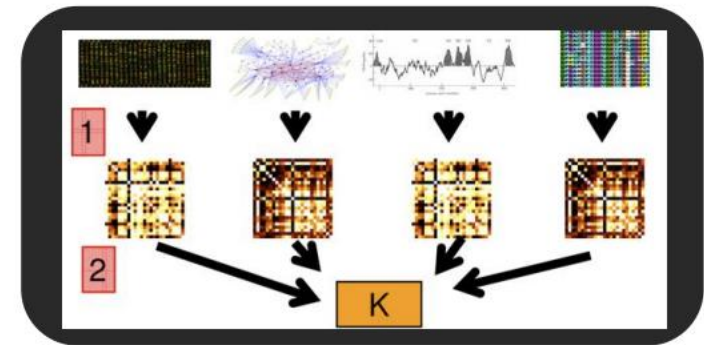


Core challenge 3: Fusion

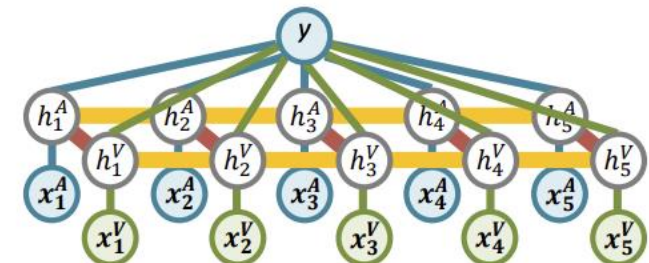
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B Model-Based (Intermediate) Approaches

- 1) Deep neural networks
- 2) Kernel-based methods
- 3) Graphical models



Multiple kernel learning

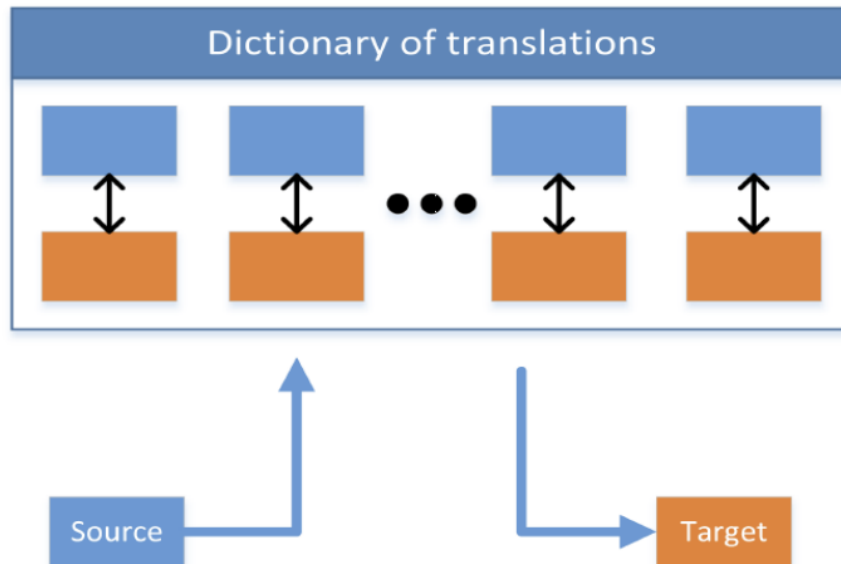


Multi-View Hidden CRF

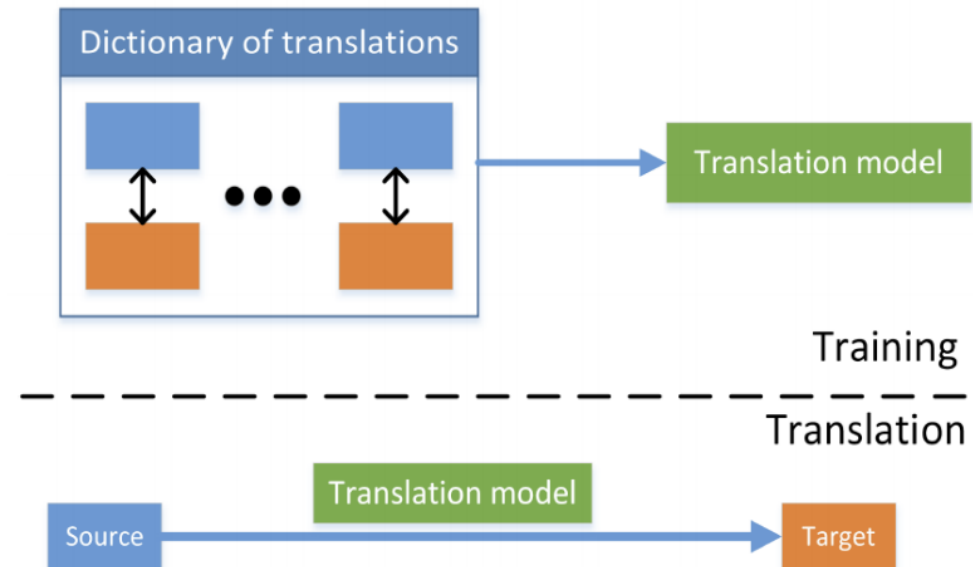
Core challenge 4: Translation

- **Definition:** Process of changing data from one modality to another, where the translation relationship can often be open-ended or subjective

A Example-based



B Model-driven



Core challenge 4: Translation



Visual gestures
(both speaker and
listener gestures)

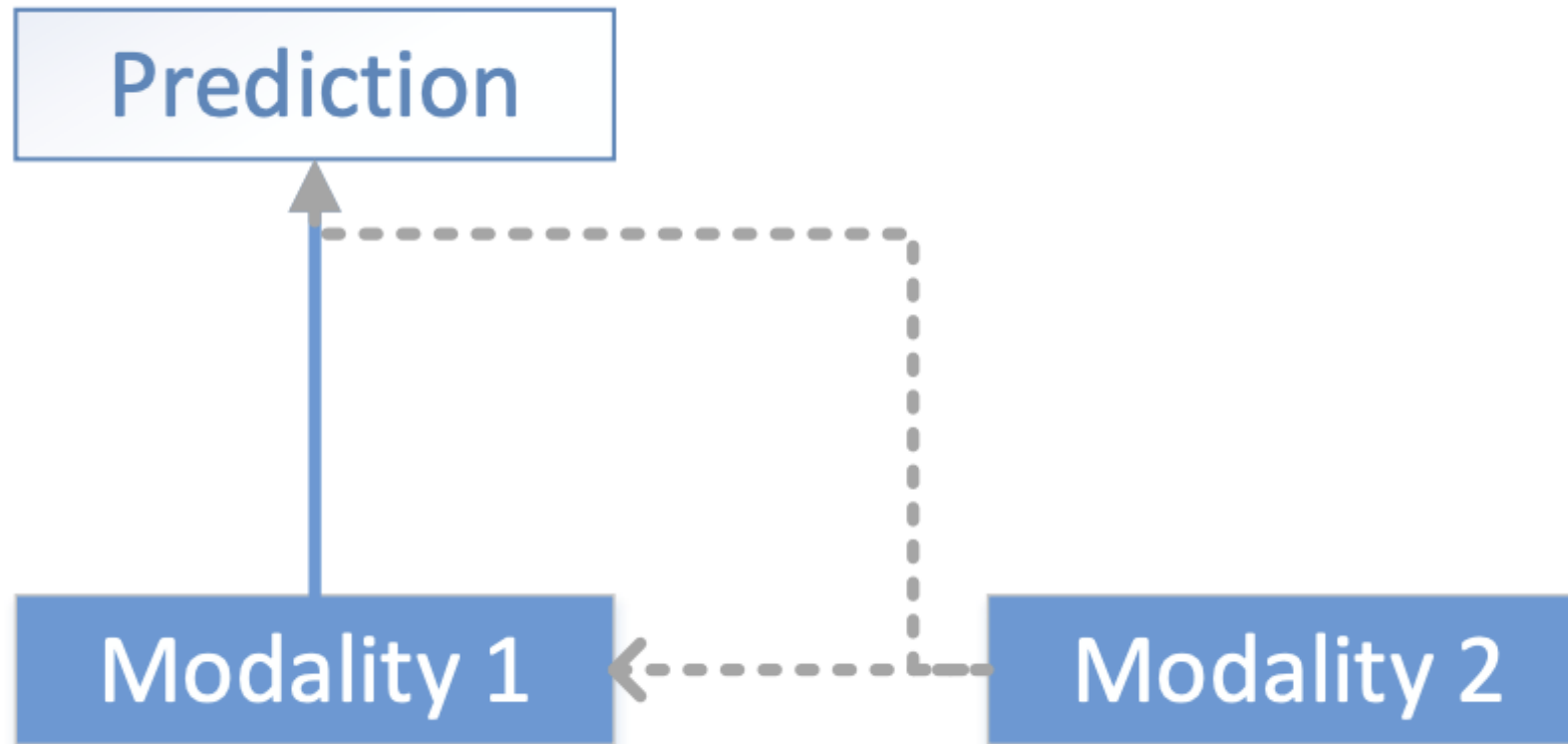


Transcriptions
+
Audio streams

Marsella et al., Virtual character performance from speech, SIGGRAPH/Eurographics Symposium on Computer Animation, 2013

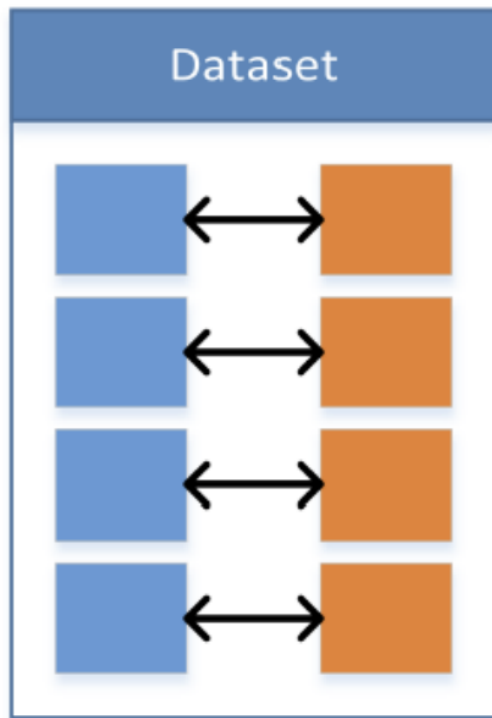
Core challenge 5: Co-Learning

- **Definition:** Transfer knowledge between modalities, including their representations and predictive models.



Core challenge 5: Co-Learning

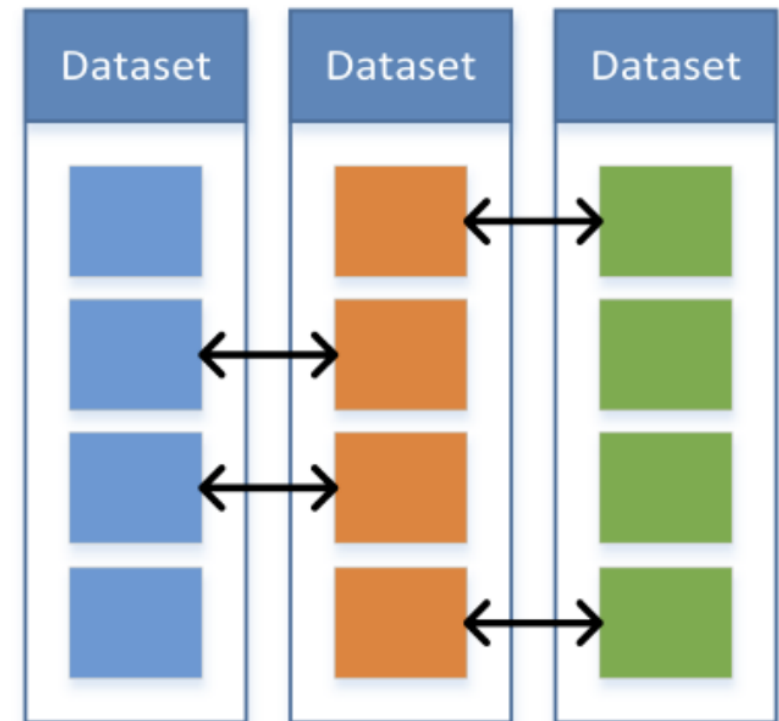
(A) Parallel

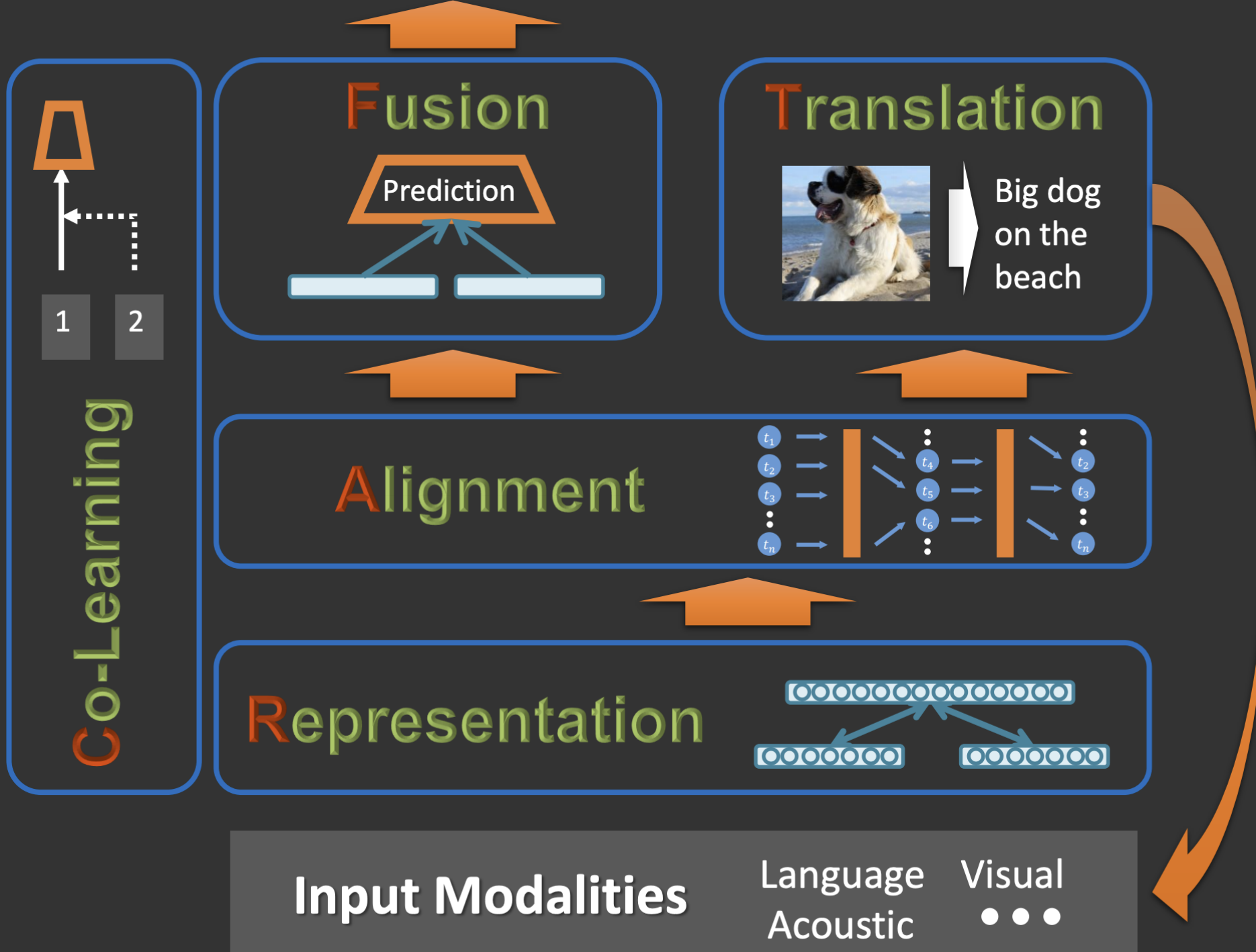


(B) Non-Parallel



(C) Hybrid





Taxonomy of multimodal research

Representation

- Joint
 - *Neural networks*
 - *Graphical models*
 - *Sequential*
- Coordinated
 - *Similarity*
 - *Structured*

Translation

- Example-based
 - *Retrieval*
 - *Combination*
- Model-based
 - *Grammar-based*

- *Encoder-decoder*
- *Online prediction*

Alignment

- Explicit
 - *Unsupervised*
 - *Supervised*
- Implicit
 - *Graphical models*
 - *Neural networks*

Fusion

- Model agnostic
 - *Early fusion*
 - *Late fusion*
 - *Hybrid fusion*

Model-based

- *Kernel-based*
- *Graphical models*
- *Neural networks*

Co-learning

- Parallel data
 - *Co-training*
 - *Transfer learning*
- Non-parallel data
 - *Zero-shot learning*
 - *Concept grounding*
 - *Transfer learning*
- Hybrid data
 - *Bridging*

Multimodal applications

	CHALLENGES				
APPLICATIONS	REPRESENTATION	TRANSLATION	FUSION	ALIGNMENT	CO-LEARNING
Speech Recognition and Synthesis					
Audio-visual Speech Recognition	✓		✓	✓	✓
(Visual) Speech Synthesis	✓	✓			
Event Detection					
Action Classification	✓		✓		✓
Multimedia Event Detection	✓		✓		✓
Emotion and Affect					
Recognition	✓		✓	✓	✓
Synthesis	✓	✓			
Media Description					
Image Description	✓	✓		✓	✓
Video Description	✓	✓	✓	✓	✓
Visual Question-Answering	✓		✓	✓	✓
Media Summarization	✓	✓	✓		
Multimedia Retrieval					
Cross Modal retrieval	✓	✓		✓	✓
Cross Modal hashing	✓				✓

Summary

- Multimodal generative AI
- Historic details of multimodal research